

Predicting the Second Gift — executive summary

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Problem

On DonorsChoose, the classroom crowdfunding platform, 71% of donors across seventeen years gave exactly once. The second gift is where retention is won or lost. A development lead at a \$500K nonprofit with no data staff can personally follow up with only a fraction of last month's first-time donors, and today she picks them by hand from gift size. Our question: at a fixed monthly capacity, which first-time donors should get the call? One of us founded GoGood Technologies, whose customers are this stakeholder; weigh our conclusions with that in mind.

Approach

We used DonorsChoose Open Data 2002–2019 (ICPSR 37898): 11.4 million donations from 3.5 million donors. Each donor gets one label, whether they gave again within twelve months of their first gift. The 2017–18 cohorts were held out and scored once, because her question is about next year's donors. Every rule is scored at her capacity, the top 10% of each month's new donors, in dollars of subsequent giving identified per contact. The baseline is her own rule, rank by first-gift size.

One finding shaped everything after it: the file holds three populations. Organizations are 0.1% of donors but 62% of repeat dollars, mostly corporate matching programs. Teachers seed their own classrooms. Pooled, any rule looks brilliant by finding the corporations. We scored citizen donors, 86% of people, the ones she actually calls.

Findings

A model beats her rule by a small, reliable margin. On the holdout, ranking by predicted second-gift value identifies **\$120 per contact**, against **\$116** for gift size and **\$21** for selecting everyone. The paired bootstrap gap is $+\$3.80 \pm 0.51$ (one standard error, 30 resamples). One call in ten reaches 58% of all subsequent giving. The model is two gradient-boosted estimators, for the chance of a second gift and its size, using first-gift attributes plus the project and school first funded. Gift size carries much of the signal; the full model adds a modest gain.

Calibration and coverage limit the result. The highest probability tenth predicts 26.6% returning and observes 24.2%. The dollar-ranked list identifies \$140 per donor in 2017 and \$103 in 2018, beating gift size in both years. Under-\$50 donors are 52.4% of the population but only **0.55% of this list**. The 14.4% figure applies to probability-only ranking, which identifies fewer dollars (\$107/contact). The two lists serve different objectives. What the model changes is who is on the list: it swaps out teacher-referred donors, friends and family recruited for one classroom who return 13% of the time, for donors who funded several classrooms at once, who return 23%.

Recommendation

Pilot the model-ranked list against gift-size ranking, measuring additional donations caused by outreach before scaling. The historical model list contains \$9.79M of subsequent giving; subtracting contact costs at an assumed \$25 leaves \$7.75M, versus \$7.44M for the gift-size list. These are **values identified less hypothetical costs, not profit estimates**. The log-amount model is a useful ranking score but is not calibrated as a mean-dollar forecast, so cost thresholds remain illustrative. Scoring runs locally without model API fees; outreach, setup and maintenance still require resources.

What we are not claiming. No causal effect: nobody was randomly assigned a call, so these are dollars *identified*, not *caused*; outreach impact remains unknown. The thank-you packet adds nothing to prediction, and it goes mostly to donors whose project got funded, so its effect cannot be separated from the project succeeding. Transfer from a marketplace to a relational small nonprofit is a hypothesis. The \$25 cost is a placeholder for a figure from real practice.

Every number reproduces with python `evals/run_all.py`; sources in `evals/results/`.